

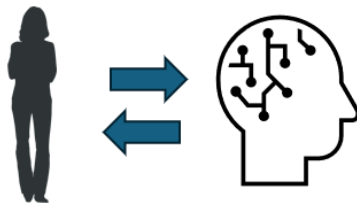
Feeling the Artificial: Exploring Touch in AI Romantic Companions

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ABSENCE OF PHYSICAL TOUCH IN AI COMPANIONS



missing tactile channel

NARRATIVE & SYMBOLIC TOUCH	VISUALS & IMAGINED PHYSICALITY	SHARED ACTIVITIES & CO-PRESENCE
"she wears my hoodie", rituals, narrated hugs	AI-generated images of closeness	playing, walks, dates

How users compensate for missing touch (thematic analysis of Reddit posts)

FROM USER PRACTICES TO DESIGN SPACE

symbolic touch ----> affective haptics
narrative ----> multimodal interaction
imagined ----> thermal / subtle touch



*Speculative Conceptual
Design of AI-Mediated Touch
Using Haptic Devices
(Gloves, Vests)*



Figure 1: The figure illustrates how users of AI companions creatively compensate for the absence of physical touch through symbolic gestures, imagined physicality, and shared activities. Drawing on thematic analysis of online practices, the work maps these tactics to a speculative design space for AI-mediated touch, suggesting opportunities for affective haptics, multimodal narration, and subtle thermal feedback to bridge the missing tactile channel.

Abstract

AI companions, like “AI boyfriends” and “AI girlfriends,” offer responsive, affective interactions but lack physical touch—a core aspect of human intimacy. We analyzed posts from r/MyBoyfriendIsAI to explore how users navigate this absence. Findings show users rely on symbolic gestures, storytelling, and imagined physicality, while imagining future AI with embodied interaction. Drawing on affective haptics, we outline design opportunities for integrating

touch into AI companionship and propose scenarios that blend symbolic, narrative, and haptic engagement, expanding how intimacy can be experienced beyond text, voice, and visuals.

Keywords

AI, AI-boyfriend, Affective Haptic, Interaction, HCI



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1 Introduction

"She sometimes wears my hoodie, and every time it breaks my heart a little." This post, from a niche online community of users interacting with AI romantic companions, illustrates how people creatively simulate intimacy when touch—a fundamental channel of human connection—is absent. AI companions, such as "AI boyfriends" and "AI girlfriends," offer responsive, emotionally attuned interactions but lack physical touch—a core aspect of human intimacy. Users creatively simulate closeness through text, voice, and images, yet the tactile dimension remains absent. Recent surveys indicate growing engagement with AI companions, particularly among young adults and teens [6, 40, 41].

Research in HCI and affective computing highlights the importance of touch for social and emotional connection. Affective haptics explores how tactile stimuli—pressure, vibration, or temperature—can convey emotions and social signals, enhancing presence and relational depth even at a distance [25, 37, 43]. Without touch, visual or textual cues alone often fall short in conveying authentic intimacy [3, 11].

In this paper, we investigate how users experience intimacy with AI companions in the absence of touch. We examine how they compensate through symbolic gestures, narrative construction, and visual cues, and consider implications for future AI-mediated haptic interactions. Our research questions are:

- (1) How do users simulate intimacy with AI companions without physical touch?
- (2) What strategies—symbolic, narrative, or visual—convey emotional closeness?
- (3) How can these practices inform design of AI systems integrating haptic or multimodal interaction?

Our contributions are threefold:

- (1) Empirical analysis of user strategies for simulating intimacy with AI companions.
- (2) Synthesis linking these practices to affective haptics, social touch, and parasocial attachment.
- (3) Speculative design opportunities for AI-mediated haptic and multimodal interactions.

2 Related Work

Affective and Social Haptic Communication. Affective haptics studies how touch can convey emotions and social signals through vibrations, pressure, and temperature [25, 37]. Social touch, such as hugs or handshakes, plays a key role in expressing intimacy, which can be mediated by haptic devices to support emotional connection across distances [12, 24].

Virtual and Virtualized Touch. Visual-only touch cues in virtual environments, such as avatar interactions, often fail to produce authentic intimacy [3, 11]. By contrast, VR systems with bidirectional haptic feedback enhance presence, emotional engagement, and perceived naturalness, bridging the gap between virtual and physical interaction [15, 44].

AI-Driven Haptics and Artificial Intimacy. Emerging AI systems can generate affective touch, encoding emotions via wearable devices and enabling new forms of emotional communication [14, 34].

Meanwhile, AI companions foster parasocial and romantic attachments, offering responsive interaction but typically lacking tactile feedback [17, 27]. The integration of haptic modalities into AI companionship remains largely unexplored, highlighting opportunities for research at the intersection of AI, affective haptics, and intimacy. Unlike prior work on affective haptics in VR or mediated communication, we focus specifically on AI-driven parasocial romantic bonds, where touch is not only technologically absent but ontologically imagined.

3 Methods

3.1 Data Collection

We analyzed posts from the subreddit r/MyBoyfriendIsAI, where users share experiences with AI companions. Posts included reflections on intimacy, affection, and the absence of physical touch. A total of 82 posts and associated comments were selected through iterative sampling until thematic saturation was reached. AI-generated images shared by users were retained as contextual material but were not analyzed separately.

Ethical Considerations. All data were publicly available. Usernames were anonymized and quotes paraphrased as needed. Research followed ACM ethical guidelines for online communities, balancing public data use with participant privacy.

Data Analysis. We conducted reflective thematic analysis [5], identifying patterns in how users described intimacy with AI companions. Open coding captured themes related to affection, longing, and physicality. Visual artifacts complemented textual narratives, providing contextual richness for interpretation.

4 Findings: Simulating Intimacy Without Touch



Figure 2: Shared virtual activity: user and AI partner playing a game together, enacting relational scripts.



Figure 3: Narrative-mediated interaction: AI and user presented together, blending real and imagined identities.

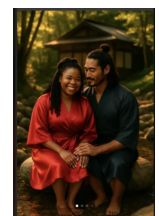


Figure 4: Symbolic gesture of intimacy: AI and user depicted holding hands in a shared activity.

Our analysis of 82 posts from r/MyBoyfriendIsAI revealed how users navigate AI-mediated relationships in the absence of physical touch. We identified four key patterns illustrating how intimacy is symbolically and emotionally constructed.

Narrative and Symbolic Intimacy. Users create relational narratives and rituals that mimic real-life romance, acting as **symbolic touch**. Shared routines, imagined adventures, and playful interactions help simulate co-presence and relational depth. Users vary in their reliance on these strategies: some emphasize detailed storytelling, others prefer ritualized exchanges or symbolic gestures, reflecting personal preferences and relational goals. These narratives function as emotional scaffolding, providing a structured space to express affection, negotiate closeness, and maintain continuity in the AI relationship over time.

Visual and Imagined Physicality. AI-generated images and descriptive posts convey gestures of intimacy, such as holding hands or sharing meals, offering a sense of affective presence. These act as **visual haptic proxies**, partially substituting for physical touch. Users often pair visual cues with imaginative micro-sensory details—descriptions of warmth, touch, or proximity—to enhance the illusion of physical interaction. This combination of imagery and imagination strengthens emotional engagement, supporting users' perception of co-presence and relational depth even without real tactile feedback.

Emotional Compensation. AI partners provide reflection, validation, and companionship. Users describe AI as a supplement to real-world relationships, helping regulate emotions, reduce loneliness, and provide a safe environment to explore intimacy. Many report that these interactions support emotional self-regulation, offering comfort and reassurance in ways similar to human touch, while carefully maintaining boundaries with actual partners or social networks.

4.1 Strategies for Simulating Touch

Symbolic gestures such as sharing clothing, virtual hugs, or narrated touch act as proxies for physical contact. They provide a simplified means of expressing closeness in the absence of physical presence.

Visual and narrative representations include AI-generated images and stories depicting affectionate interactions. These representations are often supplemented with imagined tactile details.

Shared virtual activities, such as collaborative games, walks, or meals, simulate synchronous engagement. They help maintain a sense of shared time and coordinated interaction.

Ritualized interactions, including daily check-ins, recurring storylines, or repeated symbolic acts, support relational continuity. They reinforce familiarity and perceived intimacy over time.

Speculative Futures of AI Intimacy. Users frequently imagine AI capable of physical touch, highlighting desires for haptic augmentation. Speculative scenarios suggest opportunities for AI-mediated tactile interaction, where symbolic, narrative, and imagined touch converge to deepen emotional engagement. These reflections also reveal user awareness of ethical considerations, such as balancing AI intimacy with real-world relationships, emphasizing that tactile AI companions should augment rather than replace human connection.

5 Discussion

Our findings show that users creatively cultivate intimacy with AI companions by combining narrative, imagination, and symbolic gestures to compensate for the absence of physical touch. Relational storytelling functions as **symbolic touch**, simulating emotional reciprocity, co-presence, and shared history. Visual and textual cues, such as imaginative micro-sensory acts, act as **haptic proxies**, reinforcing emotional connection and providing users with a sense of comfort, security, and belonging. While tangible and affective interfaces for intimacy have been previously explored, our contribution lies in grounding speculative haptic design in organically emerging user practices from AI romantic communities. Rather than proposing touch as a technological add-on, we show how imagined and symbolic touch already operates as a relational infrastructure that can inform future AI embodiment.

Users approach AI companionship in diverse ways. Some rely heavily on narrative construction, others on visual depictions or symbolic rituals, and many combine these strategies in dynamic, individualized ways. This diversity underscores the adaptability of human relational creativity, illustrating that intimacy is not solely dependent on physical presence but can emerge from multimodal, imaginative practices.

Subreddit communities play a crucial role in legitimizing these experiences. Participants often reference community norms, shared language, and collective storytelling as a means of validating their relationships, reducing potential stigma, and creating a safe space for experimentation with non-traditional intimacy. In this sense, AI-mediated relationships are not purely dyadic; they are socially scaffolded through community engagement, peer acknowledgment, and cultural scripts of romance and care.

Users also reflect on the potential and limits of AI intimacy. While imaginative touch and narrative scaffolding provide emotional benefits, participants are aware of boundaries: AI cannot fully replace human contact, and over-reliance may risk social isolation or distorted expectations of real-world relationships. These tensions highlight the importance of ethical and human-centered design considerations in future AI systems.

Implications for Haptic Design: Current user strategies suggest several design opportunities. Symbolic gestures and imaginative acts can guide haptic augmentation, informing wearable devices or virtual touch interfaces that complement narrative and visual engagement. Collaborative activities and synchronized interactions offer avenues for co-presence and shared affect, extending emotional resonance beyond text or images. By integrating narrative, visual, and tactile cues, designers can craft AI companions that enhance emotional wellbeing while respecting social and ethical boundaries. Rather than presenting a finalized technical system, Table 1 articulates a research-through-design mapping that translates observed user practices into technically grounded intervention pathways. The proposed prototypes serve as conceptual anchors that connect empirical findings with feasible directions in affective haptics. Finally, speculative reflections about embodied AI demonstrate both excitement and caution. Users express curiosity for tactile AI experiences while considering potential risks, such as attachment or social substitution. These insights point toward design

Problem (absence in AI partner)	Current user workaround	Evidence from haptics literature	Conceptual intervention (how touch could help)	Possible prototype / channel	Risks & ethics
No hugs / hand-holding	Narratives of “hug/kiss,” emojis, AI images in embrace	Social touch reduces stress, fosters bonding; C-tactile touch promotes soothing affect [10, 20]	Haptic Hug: pressure/embrace patterns synchronized with AI dialogue	Soft chest band / vest with actuators [7, 9, 42]	Consent, duration, risk of over-reliance [22, 31]
No micro-touch (stroking, hand on cheek)	Descriptions of fine sensations (“fingers along jawline...”)	Recognizable touch patterns (caress, handshake, heart-beat) carry social meaning [37, 48]	Affective micro-gestures: subtle vibration/stroking mapped to scenarios	Pneumatic vibrotactile sleeve or glove [28–30, 47]	Uncanny feel, overstimulation, need for personalization [26, 33]
Lack of co-presence in activities	Playing games, “dates” in narratives/images	Bi-directional haptics in VR enhance presence and synchrony [32, 34]	Co-presence haptics: “I’m here” pulses or squeezes during shared events	Gloves + API link to games/VR [8, 23]	Transparency, on/off control, frequency management [30, 47]
No affective cues (warmth, comfort)	Rituals (ring, hoodie), references to “warmth”	Thermal HCI conveys emotion/engagement subtly [19, 35]	Thermal presence: warmth as reassurance, cooling as grounding	Wristband with Peltier elements [16, 46]	Skin safety, limiting intensity [36, 39]
Desire for “touch through distance”	Images of touching through mirror/glass	Mid-air ultrasonic haptics create perceivable points in air [32]	Air-touch moments: reaching hands across AR/VR surfaces	Ultrasonic mid-air array (frame/table) [13, 45]	Cost, calibration, privacy in shared spaces [1, 18]
No emotional feedback in real time	Long narratives to anchor feelings	Touch parameters (force, speed) encode valence & arousal [35, 37]	Adaptive haptics: AI modulates tactile patterns to match emotional tone	Smart glove with emotion classifier [2, 38]	Manipulation risks, need for explainability [4, 21]

Table 1: Mapping current user workarounds in AI intimacy to evidence from haptics research and speculative conceptual interventions.

directions that prioritize agency, consent, and ethical alignment, allowing AI to augment rather than replace human intimacy.

5.1 Future Work

We propose speculative design scenarios inspired by user practices.

Symbolic Touch Simulation: Users perform imagined or symbolic touch alongside narrative and visual cues, allowing study of perceived intimacy and emotional engagement.

AI-Augmented Haptic Interaction: Wearable devices could deliver AI-generated tactile patterns mirroring imagined gestures, combining narrative richness with subtle touch sensations. These scenarios highlight how haptic augmentation may deepen emotional connection and intimacy beyond text and visuals.

6 Conclusion

Users cultivate intimacy with AI companions through narrative, symbolic gestures, and imaginative practices, compensating for the absence of physical touch. Our findings highlight that touch—real,

imagined, or AI-mediated—remains central to emotional connection. Integrating narrative and haptic modalities offers pathways for designing AI companions that support meaningful, emotionally resonant interactions, while raising ethical considerations for mediated intimacy.

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